



Energy optimization for CAN bus and media controls in electric vehicles using deep learning algorithms

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Abstract

This study offers a neural network-based deep learning method for energy optimization modeling in electric vehicles (EV). The pre-processed driving cycle is transformed into static maps and fed into a neural network for prototype energy optimization for CAN bus and media control in electric vehicles. The proposed model includes the prediction of battery state-of-charge as well as the consumption of fuel-at-destination. The controller area network (CAN) bus is the most important element in EV, ensuring its protection is the most difficult task. The abnormal messages of the CAN bus are detected using DNN. The suggested DNN model is an integrated triplet network loss which minimizes the length among the anchor sample as well as the positive sample is comparably minimum than the length measured between anchor sample and negative sample. The proposed DNN model is utilized for CAN bus and various media control in electric vehicles for effective performance.

Keywords Energy optimization · MLP · CNN · DNN · Electric vehicle · CAN bus

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1 Introduction

Vehicle electrification technologies such as unique techniques as well as a hybrid model are increasingly prominent in public as well as private transportation networks in recent times. Fossil fuels continuing usage as well as the environmental impact created by internal combustion engines (ICE) used for movement are the factors underneath it [1–4]. The main problem addressed using newly developed algorithms that integrate the performance of EV with user behavior is achieving optimum energy optimization in electric cars. The latter has a big impact on battery usage and is hard to predict. The major goal of such a field's study is to discover a way to enhance energy efficiency as well as fuel savings in HEV and EV transportation while staying within the vehicle's maximum constraints. In research, this subject is explored from many perspectives [5–11].

With the advancement of automotive electronic technologies, intelligent vehicles now contain many modern electronic components, resulting in increasingly complicated data being shuttled on the vehicle bus. Smart vehicles are moving in the direction of being unmanned as well as networked. The main future priorities of smart vehicles would be how to ensure the information security transferring across various components. The controller area network (CAN) bus of an electric vehicle is a regulated serial communicating protocol commonly utilized in internal control units of automobiles [12]. Because of its great reliability, contemporaneous performance, and convenience of wiring, this is picked by major EV manufacturers. The CAN bus collects information streams via various fundamental control systems in automobiles, including the body system, transmission system, engine, as well as other electronic devices. Broadcasting is used to communicate all branch information to the CAN bus, as well as each node may convey messages at any time to the network. It indicates that each node in the CAN network of the electric vehicle might be assaulted at any moment by fraudulent internal or external data [13]. Automobiles currently are fitted with a plethora of technological gadgets. There are sophisticated, enhanced driving supports, including parking, automated start-stop, accessory system, as well as information entertainment systems, which are linked with smart devices like smartphones, in terms of fundamental electronic controlling and media control systems.

The CAN bus aboard the vehicle may provide data to such systems. Automobiles will inevitably link to the internet facility, and such electronic gadgets and smart data systems will be a means for attackers to infiltrate the vehicle networking system, according to smart advancement. The driver might lose vehicle control if hackers gain access to such systems and link to the automobile CAN bus connection [14]. Researchers presented a study on the networking security on the market with distinct models at the Black Hat event, evaluating the capacity of various automobile manufacturers to survive malicious assaults on various models [15].

Furthermore, the Identity Document (ID) simply reflects the message's prioritization in the CAN bus system; the protocol contains neither unique address data. The retrieved electronic control unit (ECU) may not validate if the acquired data is genuine information; such that validating the received message will not be validated

using the present method, which readily contributes to the message of CAN bus forging as well as tampering by inserting fake data. This is bad, CAN bus based on a priority arbitration system allows hackers to launch denial-of-service assaults on bus communications. By listening or sniffing, an offender may repeat or overflow the vehicle bus, causing the failure of the ECU to transmit as well as retrieve data regularly. Consider dire repercussions of a smart vehicle with a full load of people driving itself if the steering controlling node is hacked. A scenario is that a malicious attacker would launch an assault on a CAN bus targeting frame, causing the user to lose control over the throttle position as well as therefore preventing the car from driving. Although not inherently harmful, a money-hungry attacker will use the vulnerabilities in the vehicle's entertainment unit to cause the vehicle to halt and show messages on the system displays. To reclaim ownership of the vehicle, the user would need to pay a bribe. As a result, for smart vehicles, establishing a contemporaneous vehicle data security monitoring system is critical [16–18].

In Sect. 1, we discussed the relevance of CAN control in vehicles and the necessity to address safety issues with energy efficiency. The paper is organized as follows: The review on energy optimization and CAN bus in the vehicle is presented in Sect. 2; A smart electric vehicle-specific anomaly detection of CAN and media control in an electric vehicle is discussed depending on the energy optimization using Deep Neural Network approach in Sect. 3; The findings of energy optimization and CAN bus control are shown in Sect. 4. Lastly, Sect. 5 includes the conclusion of the proposed work.

2 Background and related works

When looking at energy usage in EVs as well as HEVs, research on driving styles is important. It has been proposed that controllers utilize data about driving behaviors to choose the best vehicle instructions at every individual stage of the driving cycle (DC). According to findings in EV's automated control system for driving styles identification, substantial fuel savings may be achieved when the maximum aggressive drivers concentrate on reduced acceleration and the least aggressive drivers focusing on decreasing highway speeds [19, 20]. Yang et al. [21] Utilizes a machine learning technique to detect driving types (fast, medium, and conservative) as well as controllers for every driving style. Considering the aggressive approach, a sliding controller mode is recommended to increase efficacy and convenience [22]. The PI controller is recommended for modest and conservative designs.

The adaptive optimum control approach for driving style combines the identification of driving patterns with a similar consumption optimization method [23]. The pedal opening accelerator and changing rate are being evaluated using Fuzzy Logic (FL) depending on the categorization and detection of DCs. An actual vehicle test confirms the efficacy of the recommended driving type reactive control technique, demonstrating that the minimization scheme decreases usage in contrast with realistic performance. In Modern European DC, the suggested driving style identification depending on adaptive optimum control increases energy efficiency by 3.69%.

Dynamic Programming (DP) is a mathematical approach for solving complicated issues that are divided down into smaller subproblems depending on a recurring formulation. DP is utilized in research that aims to improve the energy efficiency of HEVs and EVs [24, 25]. This approach is relatively costly for huge amounts of data, which is a disadvantage of these techniques. [26] Proposes a fuel efficiency method for HEVs depending on an enhanced dynamic programming method, which effectively reduces the run time. The suggestion given in [27] is another possibility for DP.

This method uses iterative dynamic programming to address the EV eco-driving controlling issue by optimizing controlling profiles in the iteration process; moreover, owing to the high computing effort, it is not employed for long durations. An approach for designing effective EV management utilizing stochastic DCs created based on a wide range of parameters including topography, location, traffic, the operator's driving habits, and the environment is described [28]. To replicate real-world events for a vehicle, therefore, a substantial amount of relevant DC data is necessary [29]. Salameh et al. [30] presents alternate research in which DC selects data clustering as well as illustrative spots in a torque-speed profile to save computing costs.

Intelligent automobiles should go in the path of unmanned data and network connectivity in the coming future. The in-vehicle data has to be a mix of several sensors' multi-node convergence. Furthermore, the routine functioning of self-driving automobiles is impeded by data protection in the vehicle. As a result, establishing an in-vehicle data anomaly detection mechanism is critical. A unique entropy-based threat identification approach in vehicle systems is given in research [31], which uses entropy to provide an uncertainty metric of data collective items. To identify anomalies, the acquired messages of the CAN bus have been treated earlier, data entropy computation, as well as relevant length measurements, being contrasted to base sampling library generated in the estimation step. The alert will ring when there are any anomalies. If not, there would be no assaults on the CAN bus networks at this time. To identify anomaly messages in the networked vehicle, a structured method [32] is developed, that enables the detection of attacks while the vehicle is in action avoiding creating false positives.

Machine learning has got a lot of press for its usage in anomaly detection in vehicle data safety. Deep learning approaches have been utilized in a range of anomaly prediction jobs due to their capacity to autonomously retrieve data characteristics, contemporaneous learning parameters, and good accuracy rate. A new intrusion identification system is described in the research [33] employs a deep neural network (DNN) for training weight as well as retrieve the vector feature from actual CAN information packets. DNN gives the likelihood of each categorization to distinguish normal as well as attack packets while contemporaneous data packets are retrieved, allowing sensors to detect any malicious assaults on cars. Furthermore, the research [34, 35] establishes a DNN architecture that is trained on ordinary information to acquire higher-level characteristics and further utilized in forecasting forthcoming values.

3 Proposed methodology

Significant deep learning advancements are realized in automated vehicle applications, with the main elements comprising perceptual modules regulated by underlying deep Neural Networks (DNNs). DNN system collects information from a variety of sensors such as cameras, infrared sensors, light detection, and range sensors whereas outcome the knowledge needed to safely navigate a vehicle in specific situations. NNs are extensively employed in a variety of transportation applications, in addition to perception applications in automated vehicles. Deep neural networks, for example, are employed to forecast individual driver speed features, whereas NN-based models have been used to estimate traffic speed on particular road sections. To enhance vehicle energy optimization, an NN-based model is utilized for the online detection of road conditions as well as traffic congestion rates. For electric vehicle energy optimization, a deep NN approach is utilized. Several NN structures are created and utilized to accurately predict the real EV battery charging SoC and to control various media and CAN bus in an electric vehicle.

3.1 Energy optimization using deep neural network

Electric vehicle energy optimization modeling depending on deep NNs as well as actual driving cycle characteristics as inputs is not explored. This study presents a unique data-driven method to electric vehicle energy optimization modeling using neural networks that bridge the gap. If extensive driving cycle databases are accessible, such as about vehicle fleets supplied by GPS/GPRS tracking devices, the technique is especially well suited. To render them suitable used as input data to neural networks, then a preprocessing approach for converting distance-varying and time-varying driving cycles into 1-dimensional or 2-dimensional static maps is presented. Numerous deep feed forwarding artificial neural network designs, comprising multilayer perceptron's (MLPs), CNN's, as well as three alternative model input types, are evaluated and analyzed as illustrated in Fig. 1. An extended range EV method generates the energy optimization data to efficiently drive the vehicle. Depending on the produced information, two energy optimization models are developed to estimate: battery state-of-charge (SoC) together with fuel utilization after driving cycle at goal point, and all-electric range. The resulting models may be used in a variety of off-line energy requirement analyses and then on-line energy usage predictions and management techniques.

Electric Vehicle tracking device relying on GPS as well as GPRS technologies has been used to gather driving data. Within three months of duration, driving data are collected constantly, that is, 24 h a day. The data sample interval is set to one second. Aside from GPS information, a collection of information from the controller area network (CAN) bus of an electric vehicle is also obtained such as rotational speed of the engine and overall fuel utilization. Vehicle ID number, vehicle velocity, latitude, and longitude position of vehicle, timestamps, and altitude are all included in the GPS data. The whole recorded dataset yielded 2276 driving cycles, with every driving cycle

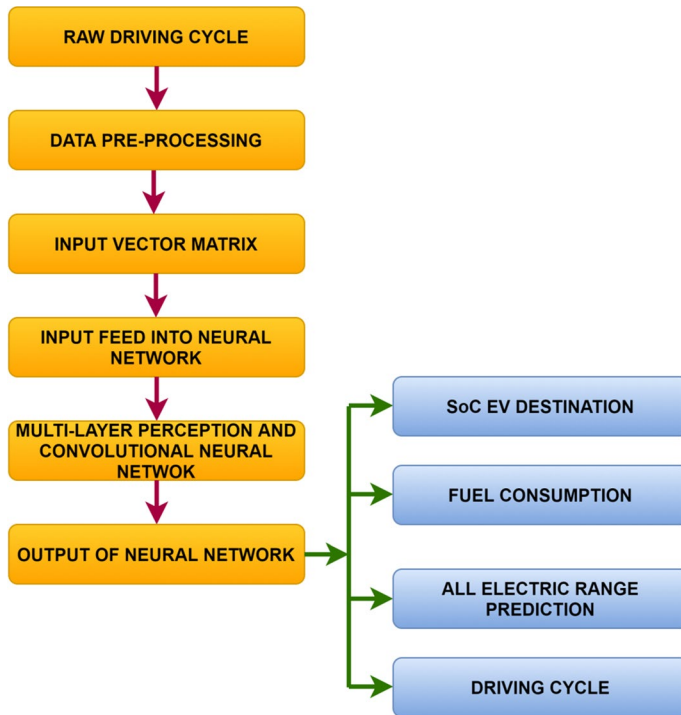


Fig. 1 Flowchart of the energy optimization in the electric vehicle using deep neural network

corresponding to a unique driving mission specified by the time gap among vehicle departure as well as return to the central distribution. It is necessary to comprehend traction control and the energy usage keep flowing from the battery approaching the wheels of the electric vehicle to offer an optimization method for driving conditions to achieve optimum efficient energy. The power requirements and limits of the EV battery discharge during functioning are also considered.

3.2 Driving cycles

The following formulas explain the longitudinal dynamics of electric vehicles:

$$\tau_L = r \left[m_v \frac{dv_v}{dt} + m_v g \sin \alpha + m_v g R_o \cos \alpha + 0.5 \rho_{\text{air}} C_d A_f v_v^2 \right] \quad (1)$$

$$\omega_L = \frac{v_v}{r} \quad (2)$$

Here, the total amount of wheel torque is signified as τ_L , vehicular mass is signified as m_v , g signifies gravitational acceleration, angular velocity is signified as ω_L , the vehicle velocity is signified as v_v , the effective tire radius is signified as r . α

signifies road gradient, and R_o signifies the rolling tire's coefficient. The standardization numerical value of air density (ρ_{air}) is considered as 1.214 kg/m^3 , C_d represents the coefficient of the aerodynamic drag, as well as A_f represents electric vehicle cross-sectional area. On wheels, the required energy may be computed as follows:

$$E_{\text{wheel}} = \int_{t=0}^{T_f} \omega_L \tau_L dt, \quad (3)$$

Here, the cyclic duration of vehicle driving is denoted as T_f . It may be stated that the following are major variables in terms of energy optimization: velocity of EV (v_v), road slope (α), acceleration ($a = \frac{dv_v}{dt}$), as well as mass (m_v). While producing inputs to the studied NNs, both vehicle velocity v_v then acceleration $\frac{dv_v}{dt}$ have been taken into consideration in this study.

Pre-processing is required in the driving cycle because it might vary concerning time and distance traveled, while feed forwarding NNs are meant to be supplied via static stimuli of fixed dimensions. Since various NN designs necessitate distinct input forms, three distinct input types (IT) are suggested, designated IT1, IT2, as well as IT3. IT1 is a 1D vector matrix of counted velocity phases in 0.5 km/h resolution in the interval of $0\text{--}90 \text{ km/h}$. The source input IT2 is a two-dimensional matrix of counted transitions among various velocity states with the same velocity range similar to IT1 input but a 1 km/h resolution. The information on vehicle acceleration is provided in this article in a convoluted method by calculating transitions between discrete velocity measurements of EV. IT3 is a 2-dimensional matrix of measured accelerations for every velocity state described in IT1, with a scale of $\pm 1.5 \text{ m/s}^2$ as well as a resolution of 0.1 m/s^2 . The data on vehicle acceleration has been explicitly provided in this scenario. It should be observed that length traveled data is effectively adjusted for regardless of input type simply detecting the EV velocities discrete values over the chosen fixed sample duration.

The following is the method for producing inputs for energy optimization models based on NN for training, then testing: separating the total driving cycle database into two different groups, with 85% of data utilized for the NN training process and balance 15% for model testing; collecting velocity and acceleration variables of EV from the driving cycle information; Counting states, comprising acceleration and velocity in discrete value-form, transitions among states, as well as storing these in 1-dimensional or 2-dimensional static maps based on the necessary NN input type (IT1, IT2, IT3), and adding the preliminary SoC_{init} value to the specified input form.

3.3 Extended range electrical vehicle

A delivery extended-range EV is based on a specific typical mid-size truck wherein driving cycle information has been collected. Initially, a model of an electric vehicle has been created with power and torque attributes identical to those of existing conventional vehicles. The range extender concept, which consists of an internal combustion engine (ICE) as well as a generator linked in a sequential hybrid arrangement is added to the electric vehicle such as the power train framework.

A backward-looking model such as quasi-static is used with the SoC battery as the state variable, and certain power train elements patterned using 1-dimensional maps including ICE and electric machines optimum torque curvatures, battery voltage under open circuit condition based on SoC and 2-dimensional maps including ICE fuel mass flowing, electric machine efficacy, voltage based on SoC. The extended range EV may operate in two modes namely charge depleting (CD) including electric driving, charge sustaining (CS) includes the hybrid driving is used to keep the SoC battery at a predefined level by integrating the generator, ICE, and motor. All-electric range refers to the distance traveled in the CD phase before transitioning to the CS phase. A control strategy that integrates a rule-based control, which includes the SoC battery controller and engine's switching logic with an immediate optimization of corresponding fuel consumption recognized as minimum utilization strategy calculates the operation point of powertrain elements.

The energy optimization is depending on the following functional dependencies since both V_f fuel consumption and SoC-at-destination SoC_{end} are primarily dependent on SoC_{init} and then the traveling distance d :

$$V_f = f_1(SoC_{init}, d) \quad (4)$$

$$SoC_{end} = f_2(SoC_{init}, d) \quad (5)$$

Since the final network is generally computationally intensive and straightforward to preprocess, the response surface technique is ideal for this proposed method of energy optimization. The phases in the modeling process are as follows: grouping measured driving cycles through length traveled, generating and validating a standard representative synthesized driving cycle for every other set, defining levels of actual SoC, and the extended range EV model over every produced synthetic driving condition for every defined stage of initial SoC.

The universal approximation theory for NNs says that if any continuous functioning in the finite-dimensional area may be approximated to any goal level of error utilizing a feed forwarding NN with an output linear layer and one lower hidden layer has a definite set of neurons. With the aim of EV energy optimization modeling, two NN topologies are investigated. The first design is corresponding to multi-layer perceptron (MLP), has a unique input layer, followed by four completely linked hidden layers containing various neurons count, and a unique output layer as shown in Fig. 2. A unique input layer, followed by two convolutional layers together with six convolutional filters possessing 2×2 sizes, a single max-pooling layer possessing 2×2 sizes, a single flattening layer, then 2 fully connected layers containing different neuron count, and a unique output layer make up CNN, the next part of NN design as shown in Fig. 2.

The former is used since it is a typical NN structure, but the latter is used as it is shown to be extremely efficient in image classification assignments (owing to its efficacy in automated feature retrieval) and processing driving cycles may be considered i.e., 2-dimensional matrices in instances of IT2 as well as IT3.

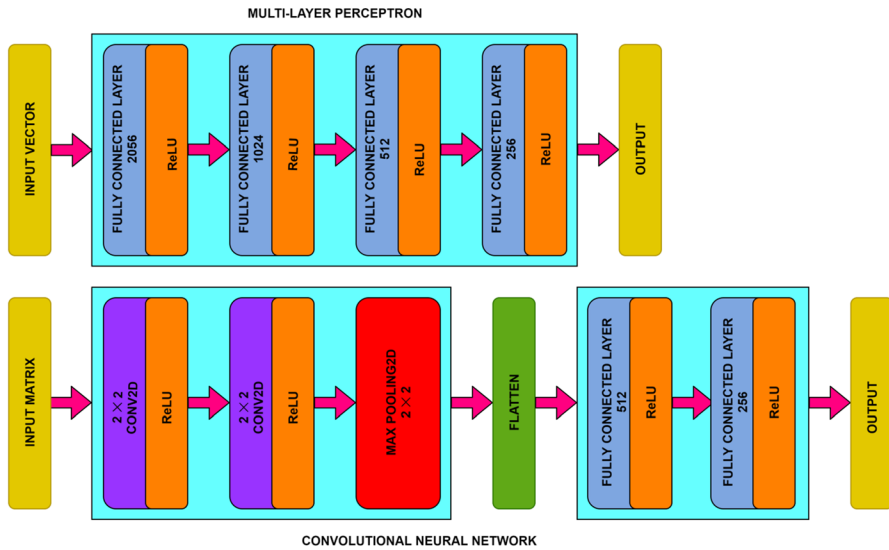


Fig. 2 Block diagram of neural network

The neural network comprising MLP as well as CNN models accept driving cycle information as inputs, including the starting value of SoC as SOC_{init} , and generate projected (V_f) fuel consumption as well as SoC-at-destination SOC_{end} . The related NN frameworks are built utilizing Tensorflow with the Keras modules as the Python backend programming language. The stride characteristic of the 2-D convolutional, as well as max-pooling layers, is fixed to 1 value. The Rectifier linear unit (ReLU) is used in the development of the DNN structure to reduce negative events to zero, allowing for a more effective network training process.

The energy optimization technique for gradient descent is utilized to train the created NNs with their parameters specified to default settings. The sample size, as well as epoch count, are both fixed at 34 and 400. The mean squared error (MSE) between the observed NN yield (i.e., y_{net}) and the desired yield (i.e., y_{train}) is expressed as an E functional loss that may be decreased via training:

$$E(X, W) = \frac{1}{2n} \sum_{i=1}^n (y_{train,i} - y_{net,i})^2 \quad (6)$$

Here, the input is denoted as X . $y_{net} = [V_f SOC_{end}]^T$ is an energy optimization vector derived from driving cycles that have been collected. The network weights are represented as W , and the entire length of the training dataset is represented as n .

The identical CNN model that predicts fuel consumption of vehicles and SoC-at-destination has been used to characterize the extended-range EV fleet but with the resultant dimension set to 1. With IT3 input, the proposed network controls each matrix row is standardized in terms of its essential components, and the vector field is dynamically raised by the relevant beginning SoC values. Because all electric range is more reliant on driving style rather than total length drove, this change to IT3 has been made.

Particularly, the regularization of matrix rows efficiently transforms the data regarding measured velocity and acceleration provinces into the transition probabilistic among the related states, leading to improve all electric-range predictions by focusing on a driving style whereas suppressing the data regarding total length traveled. In the instance of V_f as well as SoC_{end} analysis, with which the distance traveled is the most important influencing factor, the matrix rows are kept in their actual non-normalized form, representing the distance traveled.

3.4 CAN bus and media control

The entire design of CAN bus communication anomaly detection is shown in Fig. 3. The annotation CAN bus communication messages are defined as a pre-processing of the anchor during an off-line mode as well as various media control in electric vehicles based on the energy optimization. The CAN bus messages are either positive or negative. A batch is made up of the initial 3 messages of the CAN bus. Many CAN sequence data is collected throughout time, the two different sets obtained at subsequent period shall create batch together with these labeled sets, resulting in many batch sequential data. In a nutshell, the goal of this study is to detect aberrant sequences in many items. A DNN's functionality is identical to that of a convolutional neural network (CNN) in that it extracts data components. Three sequential data from the identical batch are incorporated into three physically coherent networks having weights that are shared. The retrieved 3 types of data features are then represented as three separate feature vectors. Furthermore, the triplet functional loss is utilized to estimate the similarities among random sequential data and anchor that goal is to reduce the length among normalized sequential data as well as anomaly sequential data while pushing distance among the anomalous data sequence as well as the annotated sequential data.

DNN algorithms significantly increase the identification rate in a variety of detecting activities. Data frames, distant frames, error frames, as well as overload frames, are the four categories of data frames used in a typical CAN communicating system. The lengths of these frames vary, as do their functions in the CAN communication network. The data frames out of all of these frames, perform important activities in the CAN communication mechanism. Figure 4 depicts the typical CAN packet's syntax. The initial frame, followed by arbitration field, controlling field, information field, cyclic redundancy check (CRC) field, Acknowledgment Character (ACK) field, and the conclusion of the frame are all seven-bit fields. The beginning frame comprises just a single dominant bit, signals the beginning of the data frame as well as the distant frame. Identity, as well as Radio Teletype Receiver (RTR) bits, makes up the arbitration field. This structure, the standard framing structure differs from the extended frame based on the energy optimization. The controlling field is made up of six bits, comprising

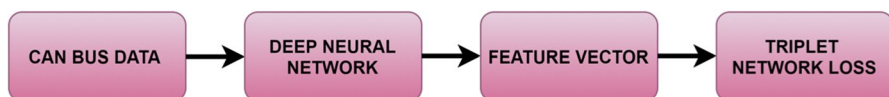


Fig. 3 CAN bus anomaly identification

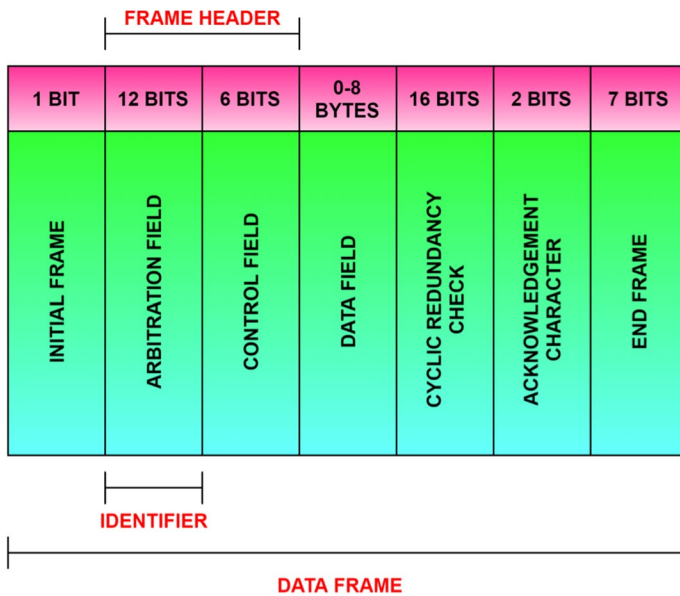


Fig. 4 Standard CAN packet structure

two reserved bits and four bits code length of data, with 0–8 bytes of informational data. The buffer is sent as per the length code, which indicates the length. A similar may be said about the information are acquired. It can range from 0–8 bytes in length; each byte that has 8 bits is the Most Significant Bit (MSB). A cyclic redundancy check (CRC) field plus CRC boundary characteristics make up the code field. The starting field, arbitration field, controlling field, data field, as well as de-filling bitstreams with 15 bits are all split polynomials in CRC computing. The ACK field consists of two recessive bits given by the transmitter that are transformed into dominant bits in the transmission node's returning interval by every node that obtains the appropriate CRC pattern.

The Controlled Area Network (CAN) bus transmits all information regarding the vehicle's status, including the deep learning approach's input characteristics. There are two tangible automobile vehicle infotainment systems. One of these is the automobile dashboard, which is located immediately in front of the driver and exhibit devices and acts as a media controller for the vehicle's functioning. The control panel provides entertainment as well as information material. It allows you to make phone calls, listen to music, etc. Each tangible automobile infotainment system is equipped with a controller that is a single-board computer. The controller dashboard can only read data acquired from the CAN bus as well as display it. The CAN bus information and the control media in EV can read and write using the infotainment controller. Thus, its display is touchscreen as it needs effective handling of the driver's input.

The proposed energy optimization using DNN contains the key data to users to influence its driving style is a contemporaneous prediction of the impending

frictional brake activation. Because the DNN can communicate with the infotainment system and automobile dashboard, the visual data may be shown in both. Though the EV dashboard is restricted by numerous rules, we opted to display the prediction on the EV dashboard since the driver must have access to every essential data. The data are instantly available to the driver concise and non-distracting manner in this manner.

4 Performance evaluation

The proposed energy optimization based on NN models is evaluated. The work area of HP Z440 together with 16 GB RAM as well as a Xeon® Intel® Processor E5-1620 v3 @ 3.50 GHz, the calculations are carried out. Initially, the collected driving cycles are analyzed, and specified input forms are acquired (IT1, IT2, as well as IT3). Following that, depending on generated extended-range EV energy optimization data, response surface, as well as NN-based models, are created, and lastly, the resultant models are assessed over such inputs, yielding numerous important performance metrics. The fundamental statistics of the generated prediction outcomes and associated residuals (i.e., differences among model-predicted as well as reference values) are reflected in the performance measures for the response surface model, whereas the NN-based models contain certain extra indicators like as: The training time duration in hours, testing time (ms), training scores corresponding to the functional loss value for the training database, testing score corresponding to the loss function value for testing database, and a total number of parameters suitable for the training process. The minimal, maximal, median, mean, as well as standard deviation of error prediction (i.e., residuals), have been considered in the basic statistics data.

The findings of NN-based models that have been shown for five distinct cases in Table 1 are contrasted to the results of response surface-based systems. Tables 2 and 3 show extensive statistical data on the forecasting outcomes for both response surface as well as NN-based systems when including (V_f) vehicle fuel consumption and SoC-at-destination (SoC_{end}). These findings show that CNN2 produces the most appropriate training as well as testing scores and the shortest assessment time. MLP1 possesses much less training time owing to the reduced input dimension, which results

Table 1 Performance evaluation of NN-based model

Model Label	MLP1 IT1*	MLP2 IT3*	MLP3 IT2*	CNN1 IT2	CNN2 IT3
Dimension of input	185	5650	8484	92×92	92×32
Training time in hour	2.3	17.9	21.8	10	8
Testing time in ms	9.3	11.03	11	5	3.3
Training value	0.01432	0.01124	0.00890	0.06457	0.00637
Testing value	0.23252	0.48467	0.34523	0.19525	0.16742

Table 2 Performance of SoC-at-Destination predictions

MODEL LABEL	MLP1	MLP2	MLP3	CNN1	CNN2
Minimal value	− 4. 2354	− 4.1242	− 4.3536	− 6.1342	− 6.3433
Median	− 0.0034	− 0.0252	0.0368	− 0.0039	0.0537
Mean	0.0036	0.0463	0.0573	− 0.0164	0.0367
Maximal value	3.8368	13.2456	10.3632	5.1346	4.2578
Standard deviation	0.5277	0.9532	0.7422	0.6258	0.5793

Table 3 Performance of Fuel Consumption Predictions

Model Label	MLP1	MLP2	MLP3	CNN1	CNN2
Minimal value	− 1.3786	− 1.1792	− 1.3778	− 2.8788	− 1.4189
Median	− 0.0218	0.0053	0.0420	0.0231	− 0.0100
Mean	− 0.0176	0.0215	0.0321	0.0213	− 0.0112
Maximal value	1.8959	3.4781	2.6891	2.8016	1.2114
Standard deviation	0.2019	0.2178	0.2571	0.2919	0.1799

in a fewer number of training parameters. The CNN2 produces the better aggregate performance regarding (V_f) fuel consumption as well as (SoC_{end}) forecasts, whereas the MLP1 outcomes simply resemble them.

We will recreate the situation of an in-vehicle network interacting in this part. CAN produce the CAN bus packet data. The training set, validation set, and testing set are the database employed. The total number of packets produced is estimated to be around 250,000. There have been 150,000 regular data of CAN bus and the remaining are aberrant data out of the overall. 70% of regular packets have been utilized for training whereas 30% of regular packets have been utilized to verify to prevent the over-fitting issue. In the data field, each data characteristic has a unique bit location. Mode data and value data may be regarded as semantic information in the proposed network framework, with mode data and value data representing a specific message relating to an ECU as well as the mode value. As illustrated in Fig. 5, the specialized command communication message acquired from mode data may control the wheels, while the command message acquired from valuable data may regulate the speed or wheel. In a short period, the modal data remains constant, but the value data may vary when the noise level changes. The useful data is utilized in the training phase, while mode data being utilized in the identification step. The data field in the CAN syntax contains 64-bit locations, in which every type of bit-distribution symbol's probability is expressed by a particular feature vector.

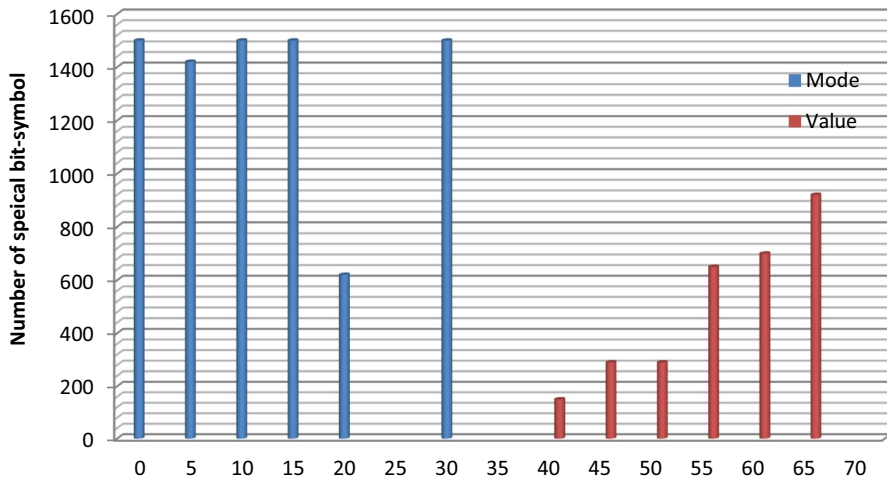


Fig. 5 Special bit-symbol Vs 8 bytes

5 Conclusion

We proposed a deep learning technique namely Deep Neural Network (DNN) in the energy optimization process in electric vehicles for CAN bus and media control. The driving distance in association with the initial SoC of the vehicle is being fed as an input to response surfaces enhances the precision rate of the proposed DNN method. Energy optimization has a great impact on vehicle controlling and therefore, Neural Network associated with driving length include other important driving cycle characteristics enclosed in Neural Network inputs via automatic feasibility analysis during energy optimization. Aside from their excellent prediction precision, NN-based models have been demonstrated to be highly computationally effective and may thus be utilized in a variety of real-time usage including vehicle media control and charge control. The proposed DNN is a unique technique for anomaly identification of messages in the CAN bus with a triplet network loss. DNN is used in training CAN bus patterns and retrieve feature vectors that correlate with CAN bus patterns in the proposed approach and ensure the controlling of various media, such as speed, torque control, CAN bus, etc., in the electric vehicle through energy optimization. Thus, the energy optimization improves the efficiency of the electric vehicle enables the user to control the various controlling media for effective performance using the Deep Neural Networks. The experimental result demonstrates the energy optimization of the proposed network model under vehicle fuel consumption and SoC-at-destination and their influence over CAN bus and media control.

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Availability of data and material Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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